



Longevity Guide: Sleep Supplements & Peptides

Evidence-Based Strategies for Better Sleep,
Recovery, and Longevity

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MEDICAL DISCLAIMER

This guide is intended for informational and educational purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider before starting, changing, or discontinuing any supplement, medication, or health regimen.

The compounds discussed—including certain peptides—may not be FDA-approved for sleep or other medical indications.

Individual responses vary, and dosages listed here reflect ranges studied in scientific literature, not personalized medical recommendations.

ABOUT THE AUTHOR



Hello there!

I'm Dr. Sahil Chopra, MD. I'm a quadruple-boarded physician specializing in Internal, Sleep, and Pulmonary/Critical Care Medicine. I completed my Internal Medicine residency at UCLA, Pulmonary/Critical Care fellowship at Loma Linda University, and Sleep Medicine training at Harvard.

We started Empower Sleep with one vision in mind: to provide Harvard-grade sleep care to as many patients as possible. The strategies I adopted in my Sleep Medicine fellowship were deeply rooted in the science and physiology of sleep science. We now apply this science here at Empower Sleep. The goal is to treat all patients as an "n-of-1" (individualized approach). Not a homogeneous, population health-based approach, which is what happens in traditional healthcare. Everyone's sleep is unique to them, as is their sleep disorder(s). When we approach every patient as an individual, we can typically get to the root cause, find a personalized solution, and have stellar clinical outcomes.

At Empower Sleep, we have cared for thousands of patients and analyzed hundreds of thousands of nights of sleep biomarkers. Our innovative care model and success is a by-product of my mentors, my amazing team, our patients, and my family.

Thank you in advance for taking the time to improve your sleep health. I hope you find this guide as a helpful tool to solve your sleep struggles.

It is my wish for all humans to live healthy and fulfilled lives.

Dr. Sahil Chopra, MD



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01 MY PHILOSOPHY

My work in sleep medicine has taught me something fundamental: **there is no single magic pill for great sleep. But there are tools that can help when used wisely and precisely.**

The supplement and peptide landscape is confusing. Social media is full of “*must-have*” products, often promoted without substantial evidence. At the other extreme, I see people dismiss supplements entirely because “*there is no proof.*”



The truth is nuanced. Some supplements and peptides have strong, peer-reviewed evidence for improving sleep quality, reducing sleep latency, or stabilizing circadian rhythms. Others are overhyped.

In my practice, I never start with supplements. I start with the **fundamentals**:

- Consistent sleep-wake schedule
- Consistent light-dark exposure (“circadian rhythm training”)
- Stress-emotional regulation
- Consistent nutrition and exercise patterns
- Ruling out obvious medical or sleep disorder (e.g. sleep apnea, Restless Legs Syndrome, adverse effects from medications, metabolic abnormalities like hyperthyroidism, etc.)

Once these are optimized and any pathology is “ruled out”, I can use **targeted supplementation** to address specific sleep challenges. This guide will walk you through my framework, including what works, what doesn’t work, and how I approach these tools with safety and precision.

02 HOW I EVALUATE SUPPLEMENTS

Every recommendation I make follows a process. If you take one thing from this guide, it is this: **supplementation should always be intentional, dose specific, and monitored to measure efficacy and/or adverse effects.**

My 3-Step Framework:

Step	Question	Examples
1 Identify the root cause of sleep disruption	What's the diagnosis or physiologic driver causing sleep disruption(s)?	Circadian rhythm misalignment, conditioned arousal, sleep apnea, RLS, metabolic dysfunction, hormone imbalance, etc.
2 Match the driver or root cause to the supplement's mechanism of action	Does the supplement's pharmacology address this?	L-Theanine for anxiety-driven sleep onset delay Magnesium for Restless Legs Syndrome (RLS) Melatonin for circadian rhythm disruption
3 Start with low dose & track	Is there a safe way to trial it with a objective + subjective feedback loop?	Track objective data and sleep architecture from wearable and/or frequent sleep testing Track subjective sleep symptoms

This process is non-negotiable. Supplements are not “*just vitamins*”—they can have potent physiologic effects, interactions with medications, and unintended side effects.

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THE FOUNDATION LAYER: ESSENTIAL NUTRIENTS

I call the following supplements “**foundation layer**” because they fill common nutritional gaps that can undermine sleep physiology.

Magnesium Glycinate or Threonate

- **Mechanism of action:** Magnesium modulates NMDA receptors and supports GABAergic activity. Both are key for initiating and maintaining sleep.
- **Evidence:** In older adults with insomnia, magnesium supplementation could improve sleep time and efficiency.
- **Dose:** 200–400 mg elemental magnesium, 30–60 min before bed.
- **Tip:** Glycinate and threonate forms are better absorbed and gentler on digestion than oxide.

Omega-3 Fatty Acids (EPA/DHA)

- **Mechanism of action:** Omega-3s reduce inflammation and modulate serotonin release. Better mood often leads to better sleep.
- **Dose:** 1–3 g combined EPA/DHA daily with meals.

Glycine

- **Mechanism of action:** Glycine can lower core body temperature, which improves sleep onset.
- **Dose:** 3 g in water 30–60 min before bed.

Vitamin D

- **Mechanism of action:** Vitamin D receptors exist in brain regions involved in sleep regulation. It influences circadian clock genes.
- **Dose:** Based on blood test; I aim for 40–60 ng/mL 25(OH)D.

Table 1 – Foundational Nutrients Summary

Nutrient	Main Effect	Typical Dose	Timing	Safety
Magnesium Glycinate/Threonate	Sleep quality, muscle relaxation	200-400 mg	30–60 min before bed	GI upset when taken in excess
Vitamin D	Circadian regulation	Personalized	AM	Hypercalcemia risk in excessive quantity
Omega-3	Mood, inflammation	1–3 g EPA/DHA	With meals	Risk of bleeding at high doses
Glycine	Body temperature	3 g	30-60 min before bed	GI upset when taken in excess



04 TARGETED BOTANICALS

The following are not “general health” supplements. They work best when matched to the specific reason your sleep is impaired.

L-Theanine

- **Mechanism:** Increases alpha brain wave activity and reduces cortisol.
- **Dose:** 200–400 mg in the evening.

Valerian Root

- **Evidence:** Modest benefit in reducing sleep latency.
- **Caution:** *Some people experience paradoxical stimulation. Very low risk of liver toxicity.*

Apigenin (Chamomile Extract)

- **Mechanism:** Binds to benzodiazepine sites on GABA-A receptors.
- **Dose:** 50–150 mg standardized extract.

Lavender (Oral Silexan®)

- **Mechanism:** Reduces anxiety and improves sleep quality.

Table 2 – Botanical Overview

Botanical	Best Use	Typical Dose	Safety
L-Theanine	Anxiety-driven insomnia	200–400 mg	Generally safe
Apigenin	Sleep onset	50–150 mg	Caution with anticoagulants
Valerian Root	Latency	300–600 mg	GI upset possible
Lavender	Anxiety + insomnia	80 mg	GI upset possible

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AMINO ACIDS & NEUROTRANSMITTER PRECURSORS

These are not “general health” supplements. They work best when matched to the specific reason your sleep is impaired.

5-HTP

- **Mechanism:** Converts to serotonin → melatonin.
- **Dose:** 100–300 mg in the evening.
- **Caution:** Avoid if on SSRIs/SNRIs.

Tryptophan

- **Mechanism:** Precursor to serotonin; less direct than 5-HTP.
- **Tip:** Take away from high-protein meals to improve brain uptake.

Taurine

- **Mechanism:** Modulates GABA and glycine receptors, reduces excitatory neurotransmission.
- **Use:** Often helpful in stress-related insomnia.

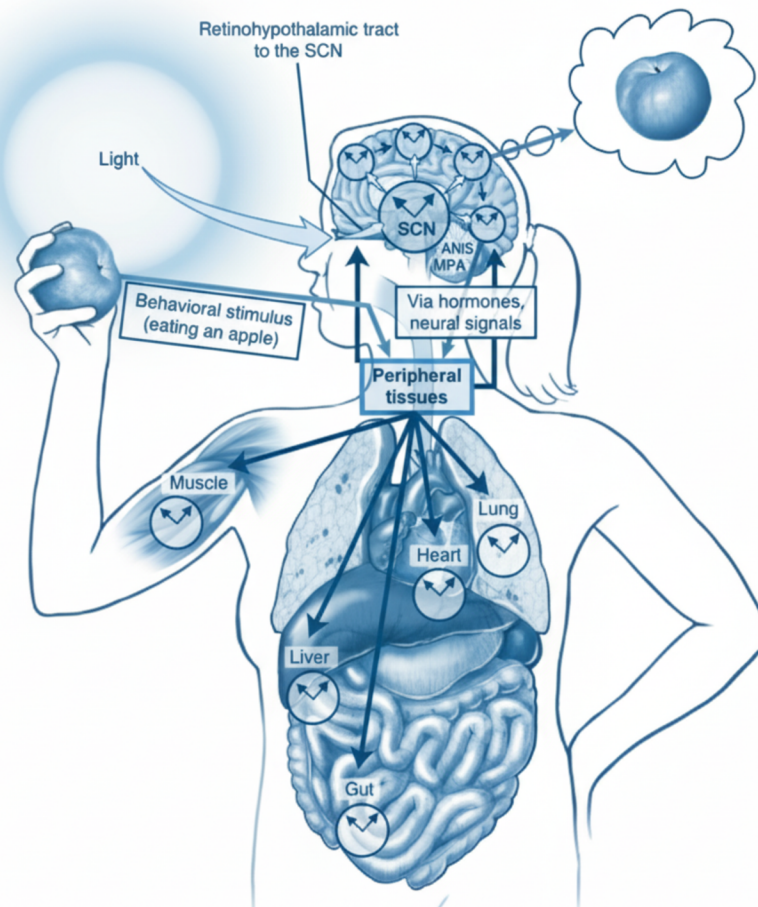
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MELATONIN DONE RIGHT

Melatonin is one of the most misunderstood “supplements.” First of all, let’s clarify that melatonin is a hormone, not a supplement. Melatonin is endogenously produced.

Normally, endogenous melatonin production is triggered every 24 hours with a specific rhythm. The presence of “light” changes this pattern by inhibiting the endogenous production. In the presence of darkness the retina sends signals to the suprachiasmatic nucleus (SCN) in the brain which tells the pineal gland, in the back of the brain, to start producing melatonin. Melatonin then goes to all the central and peripheral receptors across the body to and flips the “switch” to go from the processes of daytime to the processes of nighttime”.

- **Low or “circadian” dosing:** 0.3–1 mg mimics natural nighttime levels and is often used for jet lag.
- **High dosing (3–10 mg):** May be used in REM sleep behavioral disorder or for its sedating properties.
- **Tip:** Take 10 hours before desired wake up time.



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INTRODUCTION TO PEPTIDES

Peptides are short chains of amino acids with signaling properties. In sleep medicine, they are **not a first-line therapy**, but in select patients, may be helpful.

DSIP (Delta Sleep-Inducing Peptide)

- **Mechanism:** May promote delta-wave sleep and reduce stress reactivity.
- **Evidence:** Limited human data; more animal research.
- **Research dose:** 100–500 mcg at bedtime.
- **Caution:** Experimental, often compounded.

Epitalon

- **Mechanism:** May regulate melatonin synthesis and influence circadian clock genes.
- **Evidence:** Data from Russian longevity research; human trials limited.

Semak & Selank

- **Mechanism:** Reduce anxiety, modulate inflammation, and potentially aid cognitive recovery.
- **Sleep role:** Indirect—by lowering pre-sleep hyperarousal.

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PEPTIDE DOSING TABLE

Peptide	Mechanism	Research Dose	Timing	Safety
DSIP	Delta-wave support	100–500 mcg	Bedtime	Limited safety data
Epitalon	Circadian + aging	5–10 mg/day × 10–20 days	AM/PM	Unknown long-term
Semax	Stress/anxiety	300–600 mcg intranasal	AM/PM	Mild nasal irritation
Selank	Stress/anxiety	300–900 mcg intranasal	AM	Similar to Semax

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SAFETY & QUALITY CONTROL

- Always use tested **third-party tested** products (USP, NSF, Informed Sport).
- Avoid “*proprietary blends*” without full ingredient disclosure.
- Be aware that peptides may require prescription or compounding.

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COMBINING INTERVENTIONS

Stacking too many supplements at once is a common mistake.

I use a **“layering” approach**: add one, test for 2–4 weeks, then consider adding another if needed.

Tracking subjective and objective metrics is critical in this process.

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CLINICAL ARCHETYPES

1. **The Overthinker:** L-Theanine + Glycine → 20 min faster sleep onset.
2. **The Shift Worker:** Melatonin microdose + light therapy → more stable circadian rhythm.
3. **The High-Stress Executive:** Lavender + guided breathing → reduced nighttime awakenings.

12 4-WEEK IMPLEMENTATION PLAN



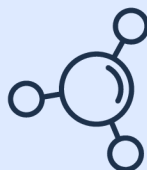
1

Optimize sleep hygiene & add magnesium



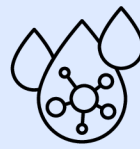
2

Add targeted botanical(s)



3

Consider amino acid/precursor



4

Evaluate peptide necessity

13 LAB TESTING FOR PERSONALIZATION

Marker	Why it Matters
Ferritin	Low iron → restless legs
25(OH)D	Vitamin D status
hs-CRP	Inflammation
Fasting glucose, insulin, TSH, free T4, sex hormones	Metabolic and hormonal health

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